

Resume to Job-Description Fit Scorer

Chetan Singh | Task 1 | AI-assisted implementation

1. Design parameter and reason. I chose a cosine-to-score range of 0.40 to 1.00, exposed in config.json. The initial 0.30-0.85 range saturated strong matches. Raising the ceiling retains differences between an exact match and a paraphrase; the 0.60-wide range maps a 0.05 cosine change to 8.33 points before clipping. These are heuristic values selected on three synthetic examples, not a validated probability scale. FastEmbed's BGE-small model runs locally. Each criterion is matched to short resume passages, then category and importance weights are normalized over the extracted criteria. Scores remain evidence for human review.

2. Failure actually observed. The first real calibration run returned A=100.00, B=99.82 and C=52.90. A/B looked consistent, but almost every strong criterion hit the ceiling, hiding meaningful differences; raw best similarities for A were approximately 0.89-0.97. The cause was the 0.85 ceiling, not parser failure. I widened the mapping and retained both result files so the change is auditable. Separately, an actual corrupt-PDF HTTP request returned 422 instead of a fit score. Model-load/inference failures return 503 rather than fabricated fallback results.

3. Metric and lesson. After the change, the same JD scored A=87.81, B=78.89 and C=31.82. The similar-resume gap was 8.92 points, within the predeclared 10-point check, and both exceeded C by at least 47.07 points. Repeating A changed its score by 0.00; one warm repeat took 112.49 ms, excluding initial model download. All 24 automated tests passed. A separate live HTTP input returned 200 and score 80.58. These measurements demonstrate a working pipeline and limited consistency, not general hiring accuracy. Raw outputs and the reproducible scripts are in results/ and scripts/.

4. Unfinished work and next step. OCR, reliable tenure arithmetic, compound-requirement splitting, robust negation handling and a recruiter-labeled benchmark are unfinished. A multi-skill bullet can still match one skill strongly; reviewers must inspect the evidence. Next I would evaluate a separate labeled set, including paraphrases, contradictions and layout changes, measure score drift and criterion agreement, then tune on a development split and report held-out results. I would also pin the model revision. I omitted deployment, authentication, a custom frontend and a database because they do not help test the assignment's core reasoning. The personal walkthrough and final public-link checks remain submission tasks.